



X-57 Electric Plane

NASA's new funny looking electric X-57 plane is carrying some amazing tech on it.
<http://digit.in/nasax57>



Venus' Electric Winds

Scientists have discovered that the electric winds on Venus stripped it of water.
<http://digit.in/elctrowind>

5 times that space totally shocked us

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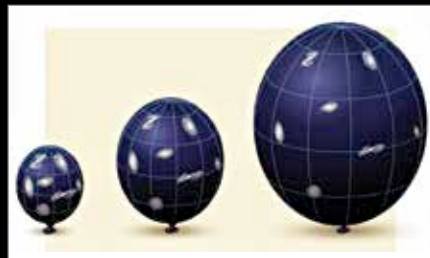
Looking up at the heavens on a moonless night, the inky blackness of the sky speckled with the few stars that light pollution still allows us to see, represents the vast depths of human ignorance. Despite the best efforts of theoretical physicists and despite what they might tell you, we are still only at the tip of the proverbial iceberg of the true nature of cosmology and astrophysics. Over the years theories have been raised, proved, rebutted, brought down and reformulated and with every successful attempt, we get a little more insight of how space works. Yet, despite this it is not the unknown that surprises us poor mortals, it is the disproving of the known, of theories that we assumed to be true; so true in fact that we often didn't even bother questioning them. Time and again, human complacency has been dealt a slap in the face by some shocking observations about space. In this article, we look at five instances where space has shocked us and given us much to ponder about.

1 The expansion of the universe

The expansion of the universe was confirmed through many astronomical observations in the early half of the 20th century. This was the period when the Big Bang hypothesis was gaining popularity, a hypothesis that stated that the universe originated from one single event, what is scientifically known as a singularity. What was surprising, however, was the observation that galaxies that are further away seem to be moving away at a faster speed. At first thought, this may seem contradictory as it seems to imply that all objects are simultaneously the centre around which the expansion occurs. This conundrum is resolved by hypothesising that it isn't galaxies as such that move away from each other, it is space itself.

and we were caught
completely unawares!

To visualise this, take a balloon and draw a few dots on the surface with a pen. Now, as you blow into the balloon, you are basically simulating the same kind of expansion that happens in the universe, where the distance between the galaxies seems to increase because the balloon itself is expanding. If we trace this back, it implies that space (and consequently time) were concepts that were created at the moment of the Big Bang. Till date, this



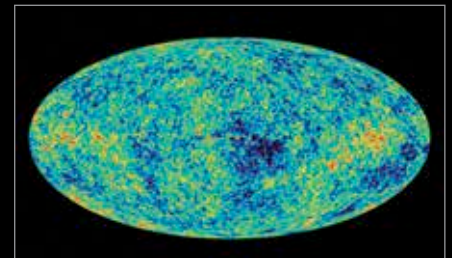
An expanding balloon showing how galaxies are moving away from each other

remains the most popular theory for the origin of the universe, but this also poses a problem as we can no longer reasonably study what happened before the Big Bang. This observation had huge impacts for cosmological models and continues to serve to remind us that the universe works in strange ways.

2 The discovery of cosmic background radiation

Around 1950, it was predicted that the entire universe is bathed in a radiation. This radiation is a remnant of the biggest cosmological event of all time: The crea-

tion of the universe itself. This radiation, called cosmic background radiation is the oldest light in the universe, dating back to the time of the recombination from Big Bang cosmology. This theoretical prediction was a massive requirement for validation of the Big Bang hypothesis. More than that, it also provides a way for us to see more in the universe, giving us a new pair of eyes with which to survey the skies. Normally, when one looks at the cosmos through a traditional telescope, the space between stars is dark. However, through a radio telescope, there is a faint glow when



A map of the cosmic background radiation in the universe

tuned to the right frequency and temperature that gives us a vast range of data that was unavailable earlier. A lot of predictions were made about the exact nature of this radiation and the properties that it would possess but the actual discovery was accidental. Working at Bell Labs in Holmdel, New Jersey, in 1964, Arno Penzias and Robert Wilson were experimenting with a 6 meter (20 ft) long, supersensitive horn antenna which was originally built to detect radio waves bounced off satellites. These radio waves they were studying were faint and required a lot of noise cancellation and a reduction of tempera-